OTH REGENSBURG

Praktikum Regelungstechnik (RTP)

Versuch T2: Systemidentifikation

Versuch 1

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`Dokument-Version: 02`

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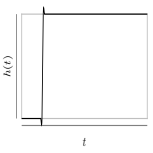
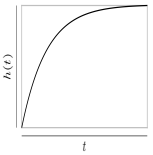
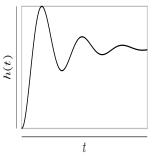
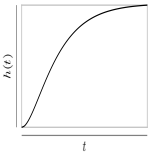
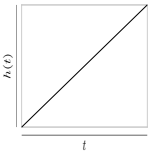
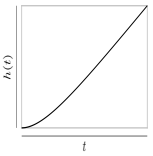
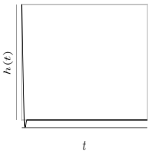
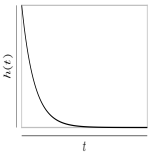
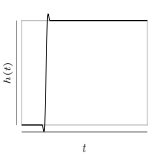
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1 Preperations

In preparation for this experiment, the files provided in ELO were downloaded and saved [**RTP˙Experiments**]. In addition, the step responses for various system behaviours were researched. As tasked in Tasks $task:a)V$ and $task:b)V$ $task:c)V$ $task:d)V$

1.1. Different basic system behaviors

Check names of coloumns again

Systemverhalten	**Übertragungsfunktion**	**Visualization**
P-Glied (K_p)	$G(s) = K_p$ (1.1)	
PT_1-Glied (K_p, T_1)	$G(s) = \frac{K_p}{T_1 s + 1}$ (1.2)	
PT_2-Glied (K_p, ω_0, D)	$G(s) = \frac{K_p}{\frac{1}{\omega_0^2} s^2 + \frac{2D}{\omega_0} s + 1}$ (1.3)	
PT_2-Glied für $D \geq 1$ (K_p, T_1, T_2)	$G(s) = \frac{K_p}{(T_1 s + 1)(T_2 s + 1)}$ (1.4)	
I-Glied (K_I)	$G(s) = \frac{K_I}{s}$ (1.5)	
IT_1-Glied (K_I, T_1)	$G(s) = \frac{K_I}{s(T_1 s + 1)}$ (1.6)	
D-Glied (K_D)	$G(s) = K_D\, s$ (1.7)	
DT_1-Glied (K_D, T_1)	$G(s) = \frac{K_D\, s}{T_1 s + 1}$ (1.8)	
T_t-Glied (K_p, T_t)	$G(s) = K_p\, e^{-T_t s}$ (1.9)	

1.1.1. P-Glied

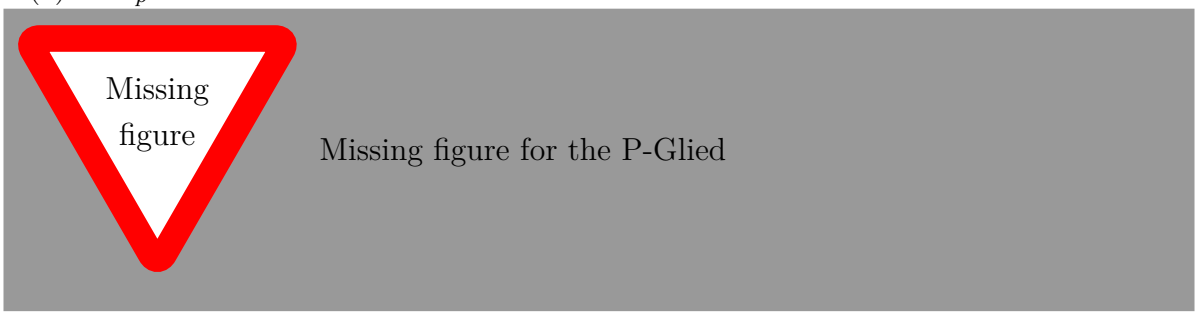
Explanation

Check explanation of the P-Glied again, look into script

Proportional-Glied, show a constant behavior in steady state for a constant input. The output is directly proportional to the input, with the proportionality constant being K_p. The equation for a P-Glied is given by

$$x_a(t) = K_p x_e(t) \tag{1.10}$$

where $x_a(t)$ is the output and $x_e(t)$ is the input. The transfer function of a P-Glied is $G(s) = K_p$ 1.1.



Parameters needed for characterization

The only parameter needed to characterize a P-Glied is the proportional gain K_p. This parameter determines the steady-state gain of the system and how much the output will change in response to a change in the input.

Step and Impulse response

The step response of a P-Glied is a step function that reaches a steady state value of K_p times the input step magnitude. The impulse response is an impulse function scaled by K_p.

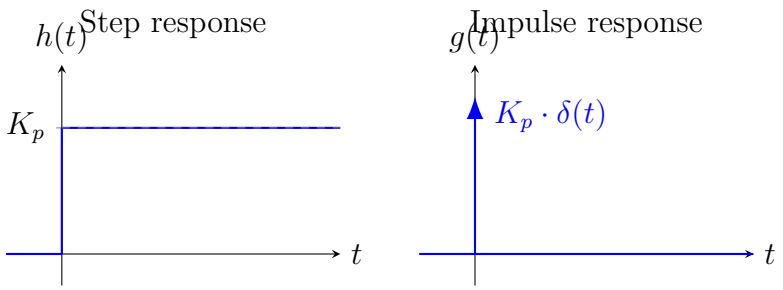


Abbildung 1.1.: P-Glied: step response (left) and impulse response (right).

1.1.2. PT$_1$-Glied

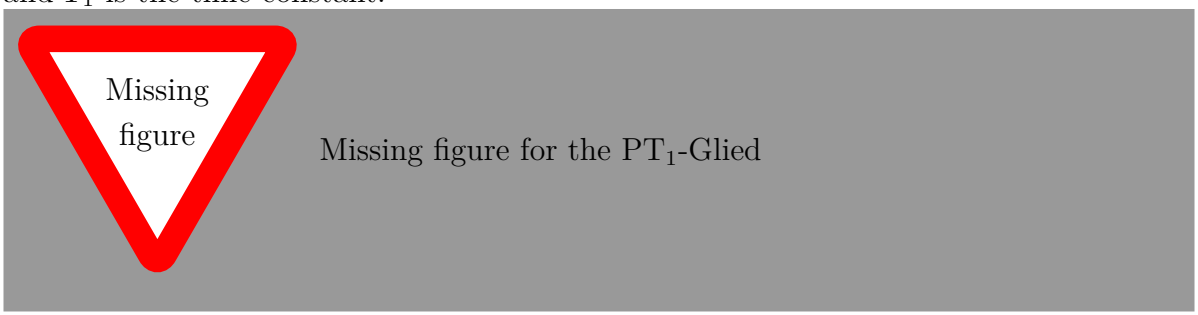
Explanation

Check explanation of the PT1-Glied again, look into script

The PT$_1$-Glied is a first-order delay system that consists of a proportional element and a first-order part. The transfer function of a PT$_1$-Glied is given by

$$T_1 \dot{x}_a + x_a(t) = K_p x_e(t) \tag{1.11}$$

or equation 1.2, where K_p is the proportional gain and the final value of the function and T_1 is the time constant.



Missing figure for the PT$_1$-Glied

Parameters needed for characterization

For the characterization of a PT$_1$-Glied, the parameters needed are the proportional gain K_p and the time constant T_1. The proportional gain K_p determines the steady-state gain of the system, while the time constant T_1 determines how quickly the system responds to changes in the input. The System is also characterized by its initial tangent, which is given by $y(t) = \frac{K_p}{T_1} t$. The point at which the initial tangent intersects the steady state value of the step response can be used to determine the time constant T_1. T_1 is the time it takes for the system to reach approximately 63% of its final value in response to a step input. The threefold multiple of the time constant T_1 at which the system reaches 95% of its final value can also be used to characterize the system's response speed.

Step and Impulse response

The step response of a PT$_1$-Glied is an exponential function that approaches a steady state value of K_p times the input step magnitude, with a time constant of T_1. The impulse response is an exponential function scaled by K_p/T_1.

1.1.3. PT$_2$-Glied

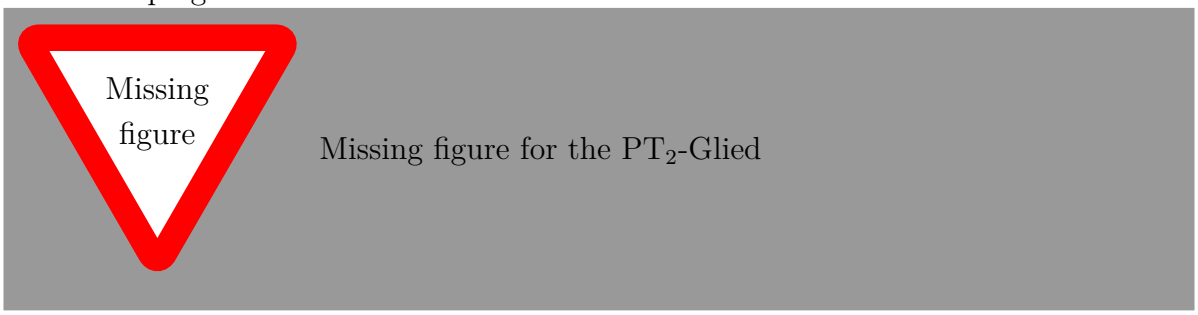
Explanation

Check explanation of the PT2-Glied again, look into script

A PT$_2$-Glied is a second-order delay system that consists of a proportional element and a second-order part, which appares delayed and oscillatory behavior becuase of its second order differential nature. The transfer function of a PT$_2$-Glied is given by

$$\frac{1}{\omega_0^2}\ddot{x}_a + \frac{2D}{\omega_0}\dot{x}_a + x_a(t) = K_p x_e(t) \tag{1.12}$$

or equation 1.3 , where K_p is the proportional gain, ω_0 is the natural frequency, and D is the damping ratio.



Missing figure for the PT$_2$-Glied

Parameters needed for characterization

For the characterization of a PT$_2$-Glied, the parameters needed are the proportional gain K_p, the natural frequency ω_0, and the damping ratio D. The proportional gain K_p determines the steady-state gain of the system, while the natural frequency ω_0 and damping ratio D determine the oscillatory behavior and response speed of the system. The damping ratio D indicates whether the system is underdamped ($D < 1$), critically damped ($D = 1$), overdamped ($D > 1$) or undamped ($D = 0$).
For an underdamped system ($D < 1$), the step response will exhibit oscillations before settling to the steady state value. The natural frequency ω_0 determines the frequency of these oscillations, while the damping ratio D determines how quickly the oscillations decay. For a critically damped system ($D = 1$), the step response will reach the steady state value without oscillating, and it will do so as quickly as possible without overshooting. It is often reached by putting two PT$_1$-Glied in series, which results in the transfer function given by

$$T_1 T_2 \ddot{x}_a + (T_1 + T_2)\dot{x}_a + x_a(t) = K_p x_e(t) \tag{1.13}$$

or

$$h(t) = K_P \left(1 - (1 + \omega_0 t)e^{-\omega_0 t}\right) \tag{1.14}$$

For an overdamped system ($D > 1$), the step response will reach the steady state value without oscillating, but it will take longer to do so than a critically damped system. The Transfer function can be rewritten as

$$h(t) = K_p \left(1 + \frac{1}{\lambda_1 - \lambda_2}(\lambda_2 e^{-\lambda_1 t} - \lambda_1 e^{-\lambda_2 t})\right) \tag{1.15}$$

where λ_1 and λ_2 are the eigenvalues of the characteristic equation of the system, given by

$$\lambda_{1,2} = -D\omega_0 \pm \omega_0\sqrt{D^2 - 1} \tag{1.16}$$

The time constants of the system can be determined from these roots, and they indicate how quickly the system responds to changes in the input. For an undamped system ($D = 0$), the step response will exhibit sustained oscillations at the natural frequency ω_0.
For the case $0 < D < 1$, the transfer function can be rewritten as

$$h(t) = K_p \left(1 - \frac{e^{-\omega_0 D t}}{\sqrt{1 - D^2}} \sin(\omega_d t + \phi)\right) \tag{1.17}$$

The Periodic time of the oscillations can be determined from the damped natural frequency

$$T_d = \frac{2\pi}{\omega_d} = \frac{2\pi}{\omega_0\sqrt{1 - D^2}} \tag{1.18}$$

The Transferfunction can be caracterized by the upper and lower envelope of the oscillations, which are given by

$$h^{\pm}(t) = K_p \left(1 \pm \frac{e^{-D\omega_0 t}}{\sqrt{1 - D^2}}\right) \tag{1.19}$$

The over swing time t_{max}, which is the time at which the first maximum of the step response occurs, can be determined from the equation

$$t_{max} = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_0\sqrt{1 - D^2}} \tag{1.20}$$

The overshoot h_{max}, can be determined from the equation

$$h_{max} = h(t_{max}) = K_p \left(1 + e^{-\pi D/\sqrt{1-D^2}}\right) \tag{1.21}$$

The normalized overshoot value $\ddot{u}$ can be determined from the equation

$$\ddot{u} = \frac{h_{max} - h(\infty)}{h(\infty)} = e^{\frac{-\pi D}{\sqrt{1-D^2}}} \tag{1.22}$$

which can be used to determine the damping ratio D from the overshoot of the step response. Given by the equation

$$D = \frac{|\ln(\ddot{u})|}{\sqrt{\pi^2 + \ln^2(\ddot{u})}} \tag{1.23}$$

Step and Impulse response

1.1.4. I-Glied

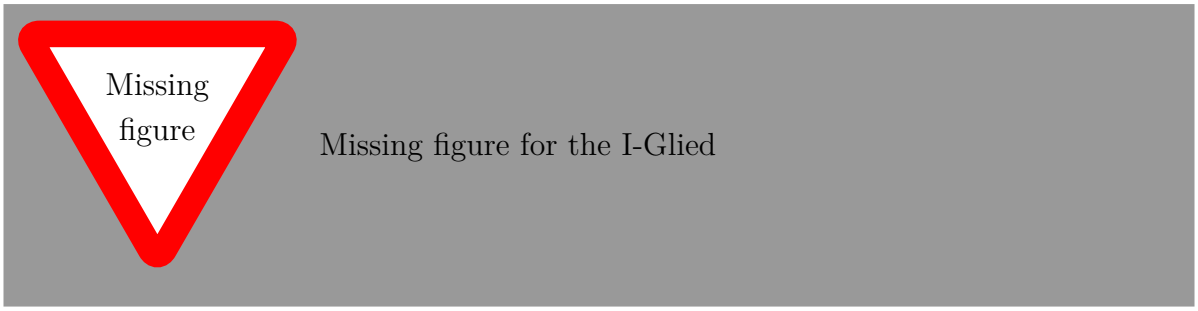
Explanation

Integral-Glied, are transfer functions that consist of an integrator of input signal, the output $x_a(t)$ is proportional to the integral of the input $x_e(t)$, with the proportionality constant being K_I. The transfer function of an I-Glied is given by $G(s) = \frac{K_I}{s}$ 1.5. given by the equation

$$x_a(t) = K_I \int_0^t x_e(\tau) d\tau \tag{1.24}$$

Equation 1.5 can also be rewritten as

$$h(t) = K_I \int_0^t \sigma(\tau) d\tau = K_I t \tag{1.25}$$



Missing figure for the I-Glied

Parameters needed for characterization

The only parameter needed to characterize an I-Glied is the integral gain K_I. This parameter determines how quickly the output will change in response to a change in the input. The step response of an I-Glied is a ramp function that increases linearly with time, with a slope of K_I times the input step magnitude. The impulse response is a step function scaled by K_I.

Step and Impulse response

The step response of an I-Glied is a ramp function that increases linearly with time, with a slope of K_I times the input step magnitude. The impulse response is a step function scaled by K_I.

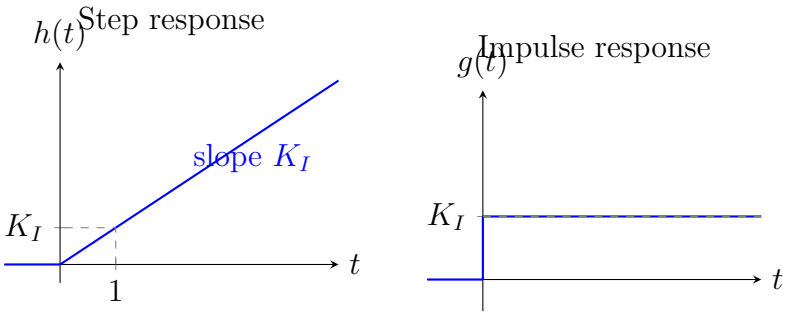


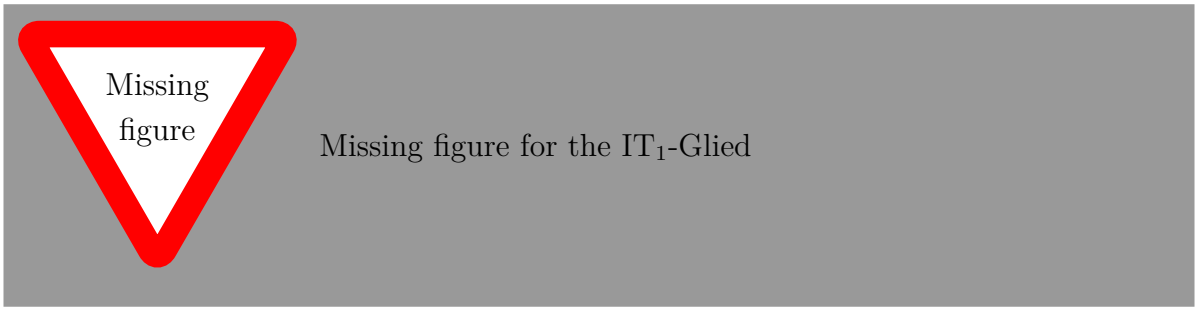
Abbildung 1.2.: I-Glied: step response (left) and impulse response (right).

1.1.5. IT$_1$-Glied

Explanation

The IT$_1$-Glied is a system that consists of an integrator and a first-order part. The transfer function of an IT$_1$-Glied is given by $G(s) = \frac{K_I}{s(T_1 s+1)}$ 1.6, where K_I is the integral gain and T_1 is the time constant. It could be seen as a combination of an I-Glied and a PT$_1$-Glied. The transfer function can be rewritten as

$$h(t) = K_I \left[t - T_1 \left(1 - e^{-t/T_1} \right) \right] \tag{1.26}$$



Missing figure for the IT$_1$-Glied

Parameters needed for characterization

For the characterization of an IT$_1$-Glied, the parameters needed are the integral gain K_I and the time constant T_1. The integral gain K_I determines proportionality of the function to the integral of the input, while the time constant T_1 determines how delayed the response of the system is to changes in the input.

Step and Impulse response

The step response of an IT$_1$-Glied is a function that increases with time, with a shape that is determined by the integral gain K_I and the delay caused by the time constant T_1. The impulse response is a function that slowly (with a delay and a shape determined by T_1) increases from zero to a the value of the integral of the input impulse scaled by K_I.

2 Experiment T2: System identification

For the 3 systems investigated in this experiment, the following paragraphs will present the experimentally identified system parameters, a documentation of the results, a real-world example of such a system, and a comparison of the given system with the ideal response of the identified system. The identifided systems are a PT1-glied for system 0 with a step-function input2.1, a PT2-glied for system 1 with a Dirac-impulse input2.2, and an IT1-glied for system 2 with a step-function input2.3.

2.1. System 0 — PT1

2.1.1. Experimentaly Identified system parameters

The experimentally identified parameters for the PT1 system are as follows:

- Proportional gain K_p: 0.999
- Time constant T_1: 1.000 s

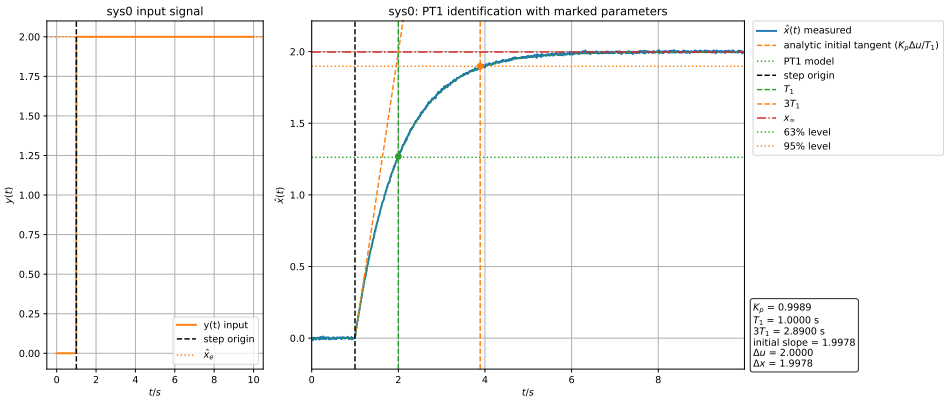


Abbildung 2.1.:

add parameters

2.1.2. Documentation of the results

Essential Commands

add essential commands for PT1 identification

Differential Equation

add differential equation for PT1 system

Transfer Function

add transfer function for PT1 system

2.1.3. Realworld Example

A practical example of such a PT1 system with step excitation is a thermometer that is suddenly placed in a warm bath. The input signal $x_e(t)$ is the bath temperature, which jumps from the ambient value to the bath value as a step at the moment the sensor is immersed. The output signal $x_a(t)$ is the temperature read by the sensor, which does not follow the bath temperature instantaneously: heat must first flow through the sensor housing into the temperature-sensitive element, so the displayed temperature rises exponentially and asymptotically approaches the bath temperature.
The proportional gain is $K_p = 1$ in this idealised case (in steady state the sensor reads exactly the bath temperature), and the time constant T_1 is set by the heat capacity of the sensor element divided by the thermal conductance between the bath and the element (housing material, wall thickness, contact area, convective heat transfer to the bath). After approximately T_1 the sensor has reached 63 % of the temperature jump, after $3T_1$ approximately 95 %. The same first-order behaviour appears in many other physical settings, for example RC low-pass filters under a step voltage or first-order mass-flow systems under a step inflow.

2.1.4. Comparison of given System with the response of the identified system

add short text about differences of the ideal system to the messured PT1 system

Deviations of the System

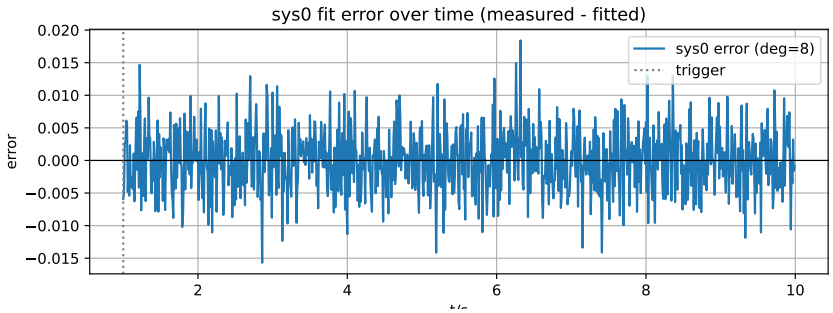


Abbildung 2.2.:

2.2. System 1

System 1 is a PT2 system, reacting to a Dirac impulse input and showing a damped oscillatory behaviour ($0 < D < 1$).

2.2.1. Experimentaly Identified system parameters

The experimentally identified parameters for the PT2 system are as follows:

- overshoot h_{max}: 28.37
- over swing time t_{max}: 0.070 s
- the damping ratio D: 0.254
- the natural frequency ω_0: 19.69 1/s
- the periodic time of the oscillations T_d: 0.330 s
- proportional gain K_p: 1.945
- input pulse area A_u: 1.000

add parameters

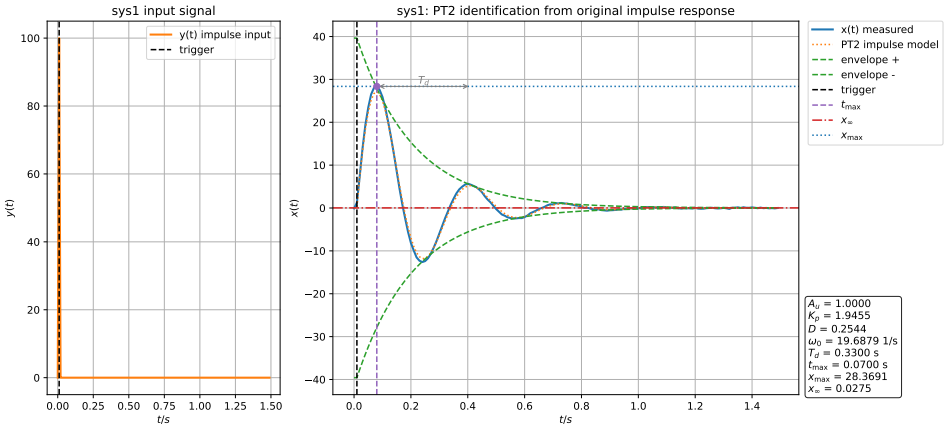


Abbildung 2.3.:

2.2.2. Documentation of the results

Essential Commands

add essential commands for PT2 identification

Differential Equation

add differential equation for PT2 system

Transfer Function

add transfer function for PT2 system

2.2.3. Realworld Example

add realworld example for PT2 system, look into script

description of the respective input and output variables.

2.2.4. Comparison of given System with the response of the identified system

add short text about differences of the ideal system to the messured PT2 system

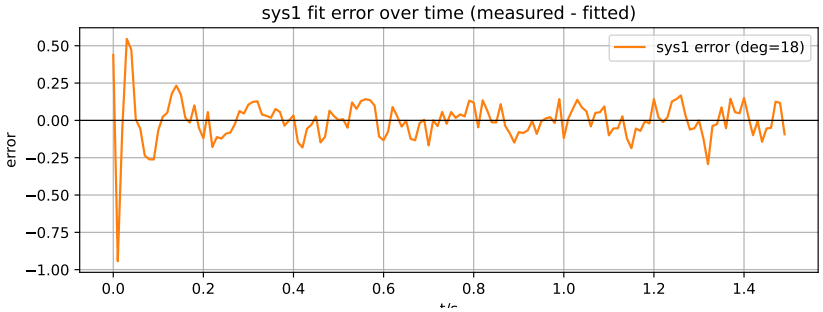


Abbildung 2.4.:

Deviations of the System

2.3. System 2

2.3.1. Experimentaly Identified system parameters

The experimentally identified parameters for the IT1 system are as follows:

- Integral gain K_I: 0.300 1/s
- Proportionalbeiwert K_p: 10 (given)
- Time constant T_1: 0.493 s

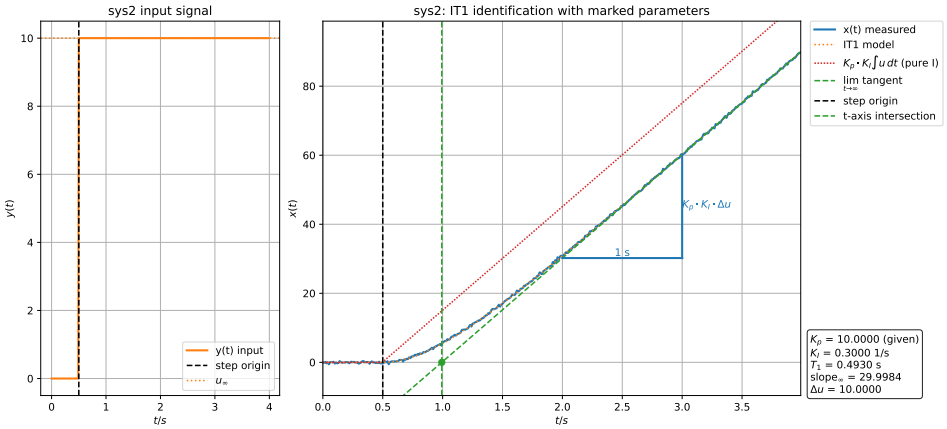


Abbildung 2.5.:

add parameters

2.3.2. Documentation of the results

Essential Commands

add essential commands for IT1 identification

Differential Equation

add differential equation for IT1 system

Transfer Function

add transfer function for IT1 system

2.3.3. Realworld Example

A clear example of an IT1 element with an upstream dead time is a water tank with a constant inflow and proportional outflow. The inflow corresponds to the input signal, and the water level to the output signal. The storage results in integrative behaviour, whilst the proportional outflow causes first-order damping — typical of an IT1 system. In the experimental setup under consideration, a dead-time element was connected upstream of the IT1 element, causing a time delay in the input signal. This means that the system output only reacts to a change after the dead time has elapsed. Such combinations of dead time and IT1 behaviour frequently occur in real-world processes, for example in sluggish piping systems or delayed control variable transmissions. description of the respective input and output variables.

2.3.4. Comparison of given System with the response of the identified system

add short text about differences of the ideal system to the messured IT1 system

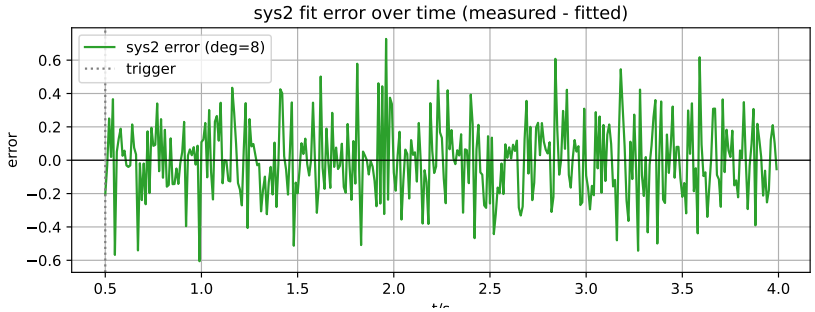


Abbildung 2.6.:

Deviations of the System

A Appendix Frontmatter

A.1. Declerations

add declerations

A.2. List of Figures

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A.3. List of Tables

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A.4. List of Variables and Constants used in this report

B Appendix Code and Resources

B.1. Code used for the analysis of the Preperations and the Experiment

add code for the analysis of the preperations, look into jupyter notebook

C Bibliography